

Mahdi Najjar

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EDUCATION

University of California San Diego

September 2022 – June 2026

Bachelor of Science. in Computer Science

Relevant Courses: Data Structures & Algorithms, Machine Learning, Databases, Operating Systems, Computer Networks, Computer Architecture, Theory of Computation, Recommender Systems

TECHNICAL SKILLS

Languages and Tools: Python, JavaScript, PHP, Node.js, HTML, CSS, C, C++, Java, Databases (SQL/MySQL), AWS, Matlab, R Studio

Developer Tools: Git, Github, Docker

Frameworks and Libraries: React, Next.js, Pytorch, Numpy, Jupyter, Pandas, Matplotlib

EXPERIENCE

Software Engineer | OPENQUANTIFY

May 2025 - August 2025

- Developed and managed RESTful APIs using Node.js and Express, re-engineering middleware and implementing strategic caching to reduce response times by 45ms and boost integration efficiency by 30% Partnered with cross-functional teams leveraging agile methodologies consistently meeting project milestones. Leveraged CS fundamentals and object-oriented programming to create scalable tools for AI driven application for 3 projects ensuring high performance and reliability

Web Data Analytics Platform | PHP, PostgreSQL, Apache

- Built a full-stack web analytics platform / stored and visualized user interaction, session, and performance data across multiple sites using JavaScript instrumentation, PHP backend services, and PostgreSQL
- Launched and maintained RESTful services on Linux cloud servers via Apache; implemented HTTPS and custom domains while improving system reliability through advanced error handling and backend testing
- Developed interactive dashboards and data visualizations with HTML, CSS, and JavaScript improving usability and real-time data interpretation

BalatJack | JavaScript, HTML, CSS

- Led a 10-member team to build modular game engine in JavaScript using object-oriented design, implementing core components (Card, Deck, Hand, Game Handler) managing game state, blind mechanics, and scoring logic. Established automated testing and CI/CD pipeline using Jest (85% code coverage), and GitHub Actions, reducing integration bugs by 40% and enforcing code quality standards across 5+ pull requests following team charter and review guidelines
- Designed and implemented interactive front-end UI components with vanilla JavaScript and CSS, improving user experience and ensuring responsive gameplay across different screen sizes

Financial Market Volatility Analysis | Pandas, NumPy, Matplotlib

- Developed a scalable data processing pipeline to analyze 60k+ social media posts from Donald Trump alongside stock and cryptocurrency datasets
- Built ETL workflows using Python (Pandas, NumPy) to ingest, clean, and normalize multi-source datasets, enabling time-series alignment across heterogeneous data streams
- Created data visualizations and statistical analyses (correlation, rolling averages...e.g.) to identify relationships between social sentiment and market volatility trends